

WESTFIELD-BARNES, MA, USA

WMO: 744915

Lat: 42.158N

Lon: 72.716W

Elev: 83

StdP: 100.34

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14775

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-14.4	-24.0	0.4	-14.7	-21.8	0.5	-12.5	11.8	-5.1	10.8	-3.8	2.1	10	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	32.9	22.5	31.3	21.8	29.6	20.8	24.3	29.7	23.5	28.3	22.7	27.1	4.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	17.7	26.2	22.2	17.1	25.6	21.4	16.2	24.9	73.7	29.9	70.4	28.3	67.3	27.1	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.5	7.6	DB	-21.8	35.8	3.2	1.7	-24.1	37.0	-26.0	38.0	-27.7	38.9	-30.0	40.1	(4)
			WB	-22.2	25.9	3.0	0.7	-24.4	26.4	-26.2	26.8	-27.9	27.2	-30.1	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-4.0	-2.4	2.2	9.1	15.0	19.6	22.7	21.7	17.7	10.8	5.0	-0.4
	DBStd	10.26	5.88	5.21	5.04	4.79	4.39	3.67	3.06	3.05	3.94	4.51	4.67	5.05
	HDD10.0	1639	434	347	247	72	6	0	0	0	1	44	163	324
	HDD18.3	3491	692	580	500	281	123	28	3	7	59	235	401	582
	CDD10.0	1573	1	1	6	45	162	288	394	363	231	71	12	2
	CDD18.3	384	0	0	0	3	21	67	139	111	39	3	0	0
	CDH23.3	3671	0	1	3	61	286	663	1336	966	331	25	0	0
	CDH26.7	1272	0	0	0	22	98	225	517	316	91	3	0	0
Wind	WSAvg	2.9	3.2	3.4	3.7	3.5	2.9	2.6	2.4	2.2	2.3	2.7	2.9	3.1
Precipitation	PrecAvg	1176	80	72	98	92	95	110	105	114	112	109	86	100
	PrecMax	1659	167	236	177	190	178	256	255	253	264	350	147	174
	PrecMin	813	34	35	19	25	27	43	50	30	30	13	16	21
	PrecStd	208	30	39	40	41	37	58	55	59	70	71	30	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	15.1	21.5	30.0	32.5	33.9	35.5	34.0	32.4	26.3	20.2	16.5
		MCWB	11.3	10.3	12.4	17.4	19.7	22.7	24.2	23.5	22.8	19.2	14.9	13.3
	2%	DB	9.2	10.5	16.5	24.0	29.5	31.4	33.1	31.7	29.1	23.0	17.6	12.3
		MCWB	6.1	6.4	9.4	13.7	19.0	21.6	22.7	22.8	20.9	17.3	13.4	9.5
	5%	DB	6.1	7.3	12.9	20.9	26.6	29.1	31.6	29.9	26.9	21.0	15.2	9.5
		MCWB	3.4	3.4	7.2	12.5	17.3	20.4	22.1	21.4	19.6	16.7	11.5	6.8
	10%	DB	3.7	4.4	10.0	17.9	23.7	27.3	29.5	28.2	24.9	18.6	12.7	7.0
		MCWB	1.3	1.4	6.0	10.9	15.9	19.4	21.2	20.8	18.8	14.5	9.3	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	12.5	14.2	18.5	22.2	24.5	25.8	25.2	24.0	21.2	17.3	13.7
		MCDB	13.0	13.8	18.9	26.8	29.6	30.8	32.4	30.9	28.8	23.7	18.8	14.9
	2%	WB	6.5	6.8	10.9	16.5	20.5	23.3	24.5	24.1	22.7	19.3	14.5	10.2
		MCDB	8.5	10.2	14.4	22.6	26.5	28.8	30.2	28.7	26.6	21.6	16.4	11.6
	5%	WB	3.7	3.7	8.3	14.2	19.1	22.2	23.6	23.3	21.6	17.3	12.1	7.0
		MCDB	5.7	6.6	12.1	18.1	24.2	27.0	28.6	27.4	25.1	20.0	14.3	9.1
	10%	WB	1.7	2.1	6.2	11.9	17.5	21.0	22.8	22.4	20.4	15.2	9.9	4.5
		MCDB	3.4	4.1	9.7	16.5	21.5	25.0	27.2	26.1	23.4	18.0	12.5	6.6
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.0	10.4	12.4	12.6	12.0	12.1	12.0	12.3	11.5	10.3	8.7
		MCDBR	11.2	13.2	15.6	18.2	17.1	14.9	14.2	13.6	14.3	14.2	13.7	11.8
		MCWBR	9.0	9.3	9.6	9.3	8.0	6.9	5.6	5.7	7.0	8.5	9.6	9.8
	5% WB	MCDBR	10.7	12.1	13.8	15.0	14.2	12.8	11.8	11.0	11.2	11.7	11.5	10.9
		MCWBR	9.0	9.1	9.6	8.9	7.4	6.7	5.4	5.2	6.2	8.3	9.4	9.8
Clear-Sky Solar Irradiance	taub	0.294	0.309	0.334	0.370	0.408	0.455	0.473	0.442	0.382	0.344	0.312	0.295	
	taud	2.427	2.394	2.418	2.365	2.295	2.194	2.151	2.226	2.371	2.491	2.536	2.517	
	Ebn at Noon	854	896	913	898	869	825	805	819	849	847	831	827	
	Edh at Noon	77	93	103	117	129	143	148	134	107	85	70	66	
All-Sky Solar Radiation	RadAvg	1.76	2.62	3.65	4.66	5.19	5.63	5.86	5.23	4.28	2.78	1.93	1.44	
	RadStd	0.14	0.23	0.27	0.50	0.56	0.54	0.35	0.28	0.33	0.28	0.17	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	+50

Nomenclature: See separate page